DOST Form 2 (for Basic/Applied Research)

DETAILED RESEARCH & DEVELOPMENT PROJECT PROPOSAL

(1) PROJECT PROFILE

Program Title:
Project Title: **SMS in Aquafeeds: Development of an Alternative Aquafeed Ingredient Using Spent Mushroom Substrate (SMS)**
Project Leader/Sex: **Liezel DC. Atole / Female**
Project Duration: **24 months**
Project Start Date: **January 01, 2025**
Project End Date: **December 31, 2026**
Implementing Agency: **College of Arts and Sciences, Partido State University**
Address: **San Juan Bautista St. Goa, Camarines Sur**
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(2) COOPERATING AGENCY/IES

National Institute of Molecular Biology and Biotechnology (BIOTECH), University of the Philippines Los Baños

(3) SITE(S) OF IMPLEMENTATION

IMPLEMENTATION SITES NO.	COUNTRY	REGION	PROVINCE	DISTRICT	MUNICIPALITY	BARANGAY
1.	Philippines	V	Camarines Sur	4	Goa	San Juan Bautista
2.	Philippines	IVA	Laguna	2	Los Baños	Batong Malake

(4) TYPE OF RESEARCH

_____ Basic
__x__ Applied

(5) R&D PRIORITY AREA & PROGRAM (based on HNRDA 2017-2022)

__x__ Agriculture, Aquatic and Natural Resources
Commodity: Aquafeeds
_____ Health
Priority Topic: _____
_____ Industry, Energy and Emerging Technology
Sector: _____
_____ Disaster Risk Reduction and Climate Change Adaptation
_____ Basic Research
Sector: _____

Sustainable Development Goal (SDG) Addressed

Life Below Water (Aquatic R&D) Agenda

(6) EXECUTIVE SUMMARY (not to exceed 200 words)

Nowadays, aquaculture has been one of the fastest growing animal production sectors in the world. The ever-increasing demand to supply the growing aquaculture industry, leads to a higher demand for fish feeds production. However, to get a good quality of feed, higher costs must be required accounting for over 50% of the production cost in fish farms (Cruz, 1996). This has led researchers to the search and development for alternative ingredient for aquafeed production. Alternative fish ingredients from locally available products, such as agricultural waste, could further provide benefits in cost savings and reduction of imports.

A good example of agricultural wastes are spent mushroom substrate (SMS), the agro-residues and the fungal mycelium left after harvesting of mushrooms. With the increased production of mushrooms, it is accompanied with the generation of tons of SMS. Due to the presence of highly significant biochemical compounds, studies have shown the potential for their use as a feed source for animals (Baek, et al 2017).

The use of SMS can help address the need for an alternative fish ingredients that is cheap and has the potential to improve the profit by incorporating agriculture waste in the formulation.

(7) INTRODUCTION

Aquaculture has contributed significantly to the country's food security, employment and foreign exchanges earnings. In the Philippines it is carried out in diverse ecosystem (freshwater, brackish water and marine). Approximately 18% of the food fish supply currently comes from aquaculture. In addition, the FAO ranks the Philippines as 11th[h] in the world in its aquaculture production as of 2019.

However, with its fast pace growth in the industry, aquaculture faces a lot of problems. Such as outbreak of diseases and pests, pollution and environmental degradation, and the high cost and poor quality of feeds (Allen, et al 2011).

Fish feeds has been the main driving force of a thriving aquaculture sector, they are essential source of nutrients and minerals, boosting the fish's overall development. The essential ingredients for the diet of fish are fish oil and fish meal, along with other important components including minerals, vitamins, cereal grains, and vegetable proteins (Cho and Kim, 2011). These are mixed together to create pellets or granules. Because of these there is a high demand for fish feeds with high and balanced nutritional diet especially protein which is an important dietary composition.

There are other feed ingredient alternatives that are more sustainably sourced. These includes plant proteins and oils from plants, yeast, insects, and algae (Ashfaq, et al 2022). For example, using oils from algae enables the nutrient requirements of the feed to be maintained without depending on fish oils. Another example is farmed seaweed that has significant growth potential as a source of fiber for aquaculture feed (Morais, et al 2020).

Due to increasing raw material and farm operating costs, this is coupled with decreasing market value for many farmed fish necessitates the farmer to reduce production costs as to maintain profitability. However, the increasing price of feed ingredients and increasing manufacturing and transportation costs were likely to have had a compound effect on global production and the price of aquafeeds (Rana, et al 2009).

Since feed represent the largest operating cost, particular attention must be focused on the development of research and farming strategies aimed in reducing feed costs and improving on-farm feed management techniques.

Another industrially developed under the agro-industrial sector is the mushroom industry, showing a projected 15% growth of demand for fresh mushrooms per year. The expansion of the mushroom industry is still increasing with the increased campaign on health, nutritional, and medicinal benefits (Zhang, et al 2014).

In the Bicol Region, mushroom production has been gaining popularity due to its high profitability and relatively low capital investments as compared to other agricultural ventures. Because of its tropical climate with high humidity and rainfall, the region is suitable for mushroom cultivation. The most commonly grown variety in the Bicol Region is oyster mushroom due to its high yield and adaptability to local conditions. Other mushrooms such as shiitake and button mushrooms are also grown, but on a smaller scale. For the production of mushrooms, the commonly used substrates are rice straw and other organic materials. The Department of Agriculture has very providing training, technical assistance, and financial support to mushroom growers. In 2016 the department has renovated the mushroom laboratory and has produced 451 mushroom mother culture and produced 1, 639 quality spawn bags each weighing 1 kilo per bag. 1,559 has been distributed to farmers and mushroom adopters and other stakeholders. With this program of the DA Region 5, it has empowered 2,093 clients through community-based trainings in mushroom production, showing a growing mushroom industry in the Bicol Region.

However, with the increased production of mushrooms in the region, it is accompanied with the generation of tons of residues referred to as spent mushroom substrate (SMS). SMS is the agro-residues and the fungal mycelium left after harvesting of mushrooms. With the increasing volume of SMS, it has faced challenges to the mushroom industry in storing and disposing the SMS.

Some studies have shown its potential for the production of fish feeds. Since the SMS content are mostly plant based that have undergo composting process, this could be utilized easier by the fish. Biochemical compounds were also detected in SMS, and showed an increase of crude protein compared from the fresh medium that is significantly an important component for fish feed production.

(7.1) RATIONALE/SIGNIFICANCE (not to exceed 300 words)

Traditionally spent mushroom substrates have been used as fertilizer for plants, and has also shown several alternative uses, such as energy production and wastewater treatment. Due to the presence of highly significant biochemical compounds, studies have also shown the potential for

their use as a feed source for animals.

SMS, as a mushroom waste compost, has been widely explored for its potential uses as feed resource for sheep, buffalo, and beef cattle due to its contents of crude protein (CP), neutral detergent fiber (NDF) and acid detergent fiber (ADF). SMS also contains a lot of mycelia, which could help improve animal antioxidant capacity and health. It was also reported that the nutritional content of the SMS varies on the type of mushroom substrate used and the type of mushroom being cultivated.

Focusing its application to aquaculture, formulation of fish feed from SMS based can also be explored. SMS contents are mostly plant based that have undergo composting process which could be utilized easier by the fish. Biochemical compounds detected in SMS, and the increase of crude protein from the fresh medium are significantly important component for fish feeds production.

(7.2) SCIENTIFIC BASIS/THEORETICAL FRAMEWORK



(7.3) OBJECTIVES

The general objective of this project is to formulate and produce an effective aquafeed in tilapia aquaculture using locally derived ingredients, the spent mushroom substrate (SMS).

Specifically, it aims:

1. To characterize the nutritional value of spent mushroom substrate.
2. To formulate and develop aquafeed using SMS as an alternative ingredient.
3. To assess the nutritional and functional properties of the developed aquafeeds.
4. To conduct feeding trial and assess growth performance in tilapia.
5. To determine cost and return analysis of the developed aquafeeds
6. To assess dietary intake of aquafeeds using metagenomics studies.

(8) REVIEW OF LITERATURE

Aquafeeds

The fast-growing economy of aquaculture leads to high demands for aquafeeds with a high and balanced nutritional diet. Asia is known to be the largest global consumer of aquafeed ingredients (FAO, 2008a). Three types of food can be used in fish ponds, this is a natural food, supplementary feeds, and complete feeds. Natural food is those that include, detritus, bacteria, plankton, worms, insects, aquatic plants, and fish that can be found in the pond. Supplementary feeds on the other hand are feeds that are regularly distributed to the fish in the pond and composed of locally available heap materials. Complete feeds may also be regularly distributed, however, these are a mixture of carefully selected ingredients to provide all the nutrients needed by the fish (Craig et al, 2017). Because of the intensive aquaculture, modern production of fish now relies on the use of artificial feeds. The use of artificial feeds helps supplement natural foods available in the pond once they become insufficient for fish. It also helps raise more fish in ponds producing higher fish yield. Artificial feeds are also advantageous for intensive aquaculture purposes due to their uniformity in their quality, nutrient balance for maximum growth effects, and its easy application.

To have a complete diet supply, all ingredients necessary for optimal growth and health should be

carefully incorporated. The common ingredients that can be found in aquafeeds are protein, carbohydrates, fats, vitamins, and minerals.

Generally, fish diets are very high in protein. Juveniles and bloodstock fish are the once that require more proteins compared to others (Tidwell & Allan, 2001). These proteins are mostly found in animal by-products, oil-bearing seeds, and processed cakes. Fish meal is commonly the major source of natural and well-balanced source of high-quality protein. Proteins are the most expensive compound in aquafeeds, thus it is important to accurately determine the exact protein requirement for each species (Cho & Kim, 2010).

Another important ingredient in aquafeeds is lipids or fats, which are high-energy ingredients providing twice the energy of proteins and carbohydrates. Lipids are typically found in fish oils that are extracted from other fish, such as sardines. Fish oils are known to be a good source of omega-3 long-chain polyunsaturated fatty acids. Fish oil is also one of the high-cost ingredients in aquaculture feed (Tasbozan & Gocke, 2017).

Another ingredient is carbohydrates such as starches, sugars, and cellulose which also provides energy to fishes. It is also used as a binding agent during feed manufacturing. Vitamins such as A, B1, B2, B3, B5, B12, Biotin, C, Choline, D3, E, Folacin, Inositol, and K are organic compounds that are also necessary for fish growth and health. Another important ingredient is minerals that are also necessary for fish to function (Kennedy, 2016).

With all the required ingredients in manufacturing fish feed, this comes at a high cost. Expensive feeds significantly affect the increased cost of production and reduced profitability of farmers. Thus, animal scientists and nutritionists are now being directed to search for cheap feed ingredients such as agricultural waste.

Alternative Aquafeed Ingredient

Due to the high cost of production of aquafeed ingredients, alternative ingredients are now being studied. The most common alternative to traditional fishmeal is plant-based products that are now widely used in aquaculture. Sources for plant-based ingredients are considered infinite, from legumes, cereal grains, oil seeds, and leaf meals. Examples of plant products that are being utilized in aquaculture are protein-rich oilseed and grain by-product meals that include soybean, rapeseed, corn gluten, wheat gluten, pea, and lupin (Tyapkova, et al 2016). Because of these advances, plant-based products make promising alternative feeds. Even though these plant-based protein sources showed lower cost alternatives, unfortunately, it has shown a highly variable nutrient profile, lacking essential amino acids, and contains anti-nutritional factors, limiting their ability to substitute fishmeal (Smith, et al 2022).

Animal sources are also being explored for their potential as an alternative feed ingredient. Meat meal, poultry meal, bone meal, feather meal, and blood meal offer good protein content for fish feeds, however, these ingredients are prone to spoilage (Carter & Hauler, 2020).

Another interesting alternative ingredient is microbial biomasses, such as microbial flakes, known as bio floc. However, because they are available in smaller quantities they are not practically relevant. Thus, single-cell proteins made up of bacteria, yeast, and microalgae are being used as an alternative ingredients Studies have shown that these organisms can have a high protein content of 45 to 65% and an amino acid profile comparable to fishmeal (Nyssolla, et al 2022).. However, this still needs to be assessed at the species level, because the absorption rates, digestibility, and bioavailability of nutrients may still vary in different microbial biomasses. Other non-conventional sources of proteins such as snails, worms, frogs, and maggots

Spent Mushroom Substrate

One of the organic wastes that have now received attention for utilization is spent mushroom substrate (SMS). This refers to composted material that has been used for mushroom cultivation, containing organic matter from agro-residues and fungal mycelium. Due to the booming industry of mushroom production, has also generated millions of tons of residues that pose a challenge for the mushroom industry in terms of storage and disposal.

Studies have shown that SMS is enriched with extracellular enzymes such as lignin peroxidase, laccase, manganese-dependent peroxidase, and other microorganisms. It also has relatively high levels of nitrogen, potassium, phosphorus, calcium, and trace elements such as silicon and iron. Because of these characteristics, SMS is commonly used as fertilizer for plants (Vrasanks, et al 2016).

Due to the presence of highly significant biochemical compounds, studies have also shown the potential for their use as a feed source for animals. These has been used as a feed source for sheep, buffalo, and beef cattle due to its content of crude protein, neutral detergent fiber, and acid detergent fiber. SMS, also

contains a lot of mycelia, which could help improve animal antioxidant capacity and health (Baek, et al 2017).

Some studies has also shown its potential for the production of fish feeds. Since the SMS content are mostly plant based that have undergo composting process, this could be utilized easier by the fish. Biochemical compounds were also detected in SMS, and showed an increase of crude protein compared from the fresh medium that is significantly an important component for fish feed production.

The paper of Pathmashi et al (2008) has showed that SMS subcultured with Pleurotus ostreatus obtained a protein value of 15.7-29%. While in the paper of Koutrotsios et al (2014), showed a protein value of 26.9-59.4% of protein value. SMS is also a good source of general nutrients such as nitrogen, potassium, phosphorus, sodium, magnesium and other trace elements. Spent mushroom also showed to contain cellulose, hemicellulose, and lignin.

(9) METHODOLOGY

Study 1. SMS Characterization and Profiling of Microorganisms

Spent Mushroom Substrate Preparation

One (1) commonly produced spent mushroom substrate will be selected and will be obtained from the Department of Agriculture – Regional Field Office No. 5 Bicol Region, Pili Camarines Sur. The substrate will be dried in an oven at 60°C for 24 h. The dried substrate will be sieved to obtain a 2 mm mesh size and will be pretreated using 1% NaOH at 90°C for 1 hour. Then, the pretreated substrate will be washed with distilled water until the pH (pH 7) and will be dried in the oven at 60°C for 24 hours. The substrates will be kept in an air-tight container for future use.

Biochemical Composition Analysis

A. *Proximate Composition, Amino Acid and Fatty Acid Profiles, and Mineral analysis*
Proximate composition, amino acid and fatty acid profiles, and mineral (P, K, Ca, Mg, Zn, Mn, Cu) analysis of SMS samples will be analyzed at BIOTECH, UPLB.

B. *Microbial Profiling*
To isolate and enumerate bacterial populations from the SMS, 100 g of samples will be placed in 900ml of 0.9% NaCl solution. The mixture will be blended thoroughly, tenfold dilution series will be prepared and 0.1ml of the suspensions will be aliquoted onto three replicate plates of Tryptic soy agar (TSA). All plates will be incubated for 24 hours at 37C. Colonies will be enumerated and populations will be expressed as colony forming units (cfu) per gram of compost.

For the microbiome analysis, bacterial genomic DNA of SMS samples will be extracted using Soil DNA Extraction Kit. The V4 hypervariable region of the 16S DNA gene will be PCR amplified using the universal primer pair 515F/806R. The purified PCR products will be pooled in equal molar concentration and sequenced on Illumina MiSeq platform following the manufacturer's instruction. Resulting sequences will be processed using a proprietary pipeline. Sequencing data will be joined and all barcodes, primers, and sequences <150 bp will be removed. Taxonomic classifications will be obtained using BLASTN against GreenGenes database (DeSantis et. al, 2006). The bacterial community composition will be measured at taxonomic levels of phylum, class, order, family, and genus. Alpha- diversity measures will be calculated within QIIME. Beta-diversity will be computed using principal coordinated analysis (PCoA) of the Bray-Curtis dissimilarity index. The functional activites of the bacterial communities will be analyzed using the open-source bioinformatics tool PICRUS (Phylogenetics Investigation of Communities by Reconstruction of Unobserved States), wherein the sequencing dataset will be normalized and the metagenomes will be predicted against the PICRUSt-formatted, characterized- protein, functional database of Kyoto Encylopedia of Genes and Genomes (KEGG).

Study 2. Production and Characterization of Aquafeeds containing SMS

Dietary Inclusion of SMS in Tilapia Feeds

Five experimental diets will be formulated with increasing inclusion levels of SMS to replace soybean meal (by weight):

T1: Control (basal diet)
T2: 25% inclusion of SMS
T3: 50% inclusion of SMS
T4: 75% inclusion of SMS

T5: 100% inclusion of SMS

Table 1. Feed formulation of the experimental diets with SMS as soybean meal replacement (g/ 100g).

Feed Ingredients	**T1**	**T2**	**T3**	**T4**	**T5**
SMS	0	8.75	17.5	26.25	35
Soybean meal	35	26.25	17.5	8.75	0
Fishmeal	16	16	16	16	16
Defatted rice bran	15	15	15	15	15
Bread flour	23	23	23	23	23
Soybean oil	8	8	8	8	8
Mineral Mix	1.5	1.5	1.5	1.5	1.5
Vitamin Mix	1.5	1.5	1.5	1.5	1.5

All ingredients will be milled and sieved prior to mixing. Dry ingredients will be mixed, followed by liquid ingredients, and will be extruded to obtain a pellet of 1.5-2mm diameter size.

Chemical Characterization of the Developed Experimental Feeds

Formulated diets will be analyzed for proximate composition. Amino acid and fatty acid profiles will be analyzed at the Feeds and Specialty Products Laboratory, BIOTECH, UPLB.

Study 3. Feeding Trial Experiment of Aquafeeds

Feeding trial experiment of the experimental SMS Aquafeeds will be conducted in Wet Laboratory of Partido State University, Sagnay Campus using an aquarium tank system. Juvenile tilapia fishes will be obtained from BFAR Region V at Fabrica, Bula, and Camarines Sur. All experiments will be performed in accordance with the guidelines of the Animal Care and Use Committee.

Acclimation and Experimental Set-up

Acquired juvenile fishes will be acclimatized in a 40-liter plastic tank with continuous aeration for 2 weeks and will be fed with basal diet. Fifteen (15) concrete tanks (1x1x1 m tank) (3 replicates for each treatment) will be prepared for the feeding trial. Tanks will be rinsed thoroughly and will be filled with tap water. The tank will be equipped with an aeration and flow-through filtration system. A total of 225 tilapia fish with an average body weight of 1.5 grams will be used in the study.

Feeding Regime and Sampling

Fish will be fed to apparent satiation twice daily. Biomass sampling will be done every 10 days for 60 days.

Growth Performance and Feed Utilization

Growth performance and feed utilization efficiency of tilapia will be assessed after 120 days of culturing period. The following parameters will be evaluated: Feed Conversion Ratio (FCR), Feed Conversion Efficiency (FCE), Feed Intake, Survival/ Mortality, and Percent weight Gain. Average body weight will be measured every 10 days. Mortalities will be noted every day before, during, or after feeding. Water quality will be monitored once a week for dissolved oxygen, temperature, salinity, and ammonia content.

Preliminary Economic Evaluation of Aquafeeds

The total cost of feeds produced per treatment will be determined based on the ingredients being used per 100 g.

Profit index will also be calculated using the formula below:

$$\text{Profit Index} = \frac{\text{Total value of fish produced} - \text{total cost of feed}}{\text{Total cost of feeds}}$$

Statistical Analysis

Data will be subjected to One-Way Analysis of Variance (ANOVA) and Kruskall Wallis Test using Statistical Package for the Social Sciences (SPSS) Version 22 to determine the effects of the different diets on the growth performance and survival of Tilapia. Duncans' Multiple Range Test (DMRT) will be used to compare the means to determine whre there is significant difference between means of the different treatments.

Study 4. Metagenomics analysis of Fish Gut Microbiome

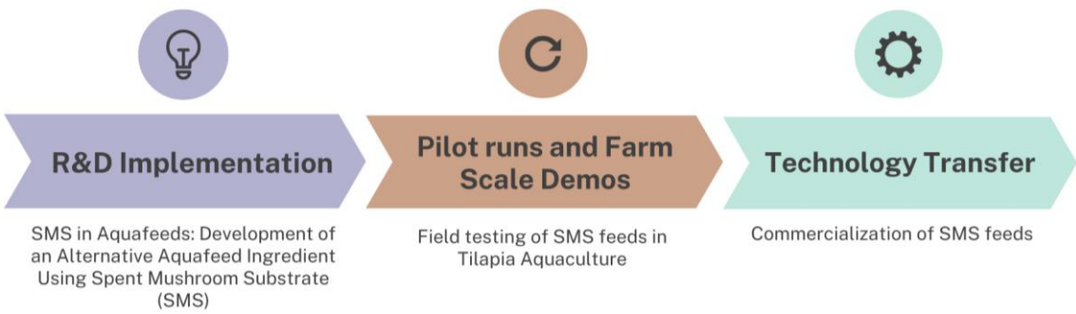
PCR Amplification and Sequencing

Bacterial genomic DNA from gastrointestinal tract of fish samples will be extracted using PowerFecal DNA kit. The V4 hypervariable region of the 16S DNA gene will be PCR amplified using the universal primer pair 515F/806R. The purified PCR products will be pooled in equal molar concentration and sequenced on Illumina MiSeq platform following the manufacturer's instruction. Resulting sequences will be processed using a proprietary pipeline. Sequencing data will be joined and all barcodes, primers, and sequences <150 bp will be removed. Taxonomic classifications will be obtained using BLASTN against GreenGenes database (DeSantis et. al, 2006).

Bacterial community composition

The bacterial community composition will be measured at taxonomic levels of phylum, class, order, family, and genus. Alpha- diversity measures will be calculated within QIIME. Beta-diversity will be computed using principal coordinated analysis (PCoA) of the Bray-Curtis dissimilarity index. The functional activites of the bacterial communities will be analyzed using the open-source bioinformatics tool PICRUS (Phylogenetics Investigation of Communities by Reconstruction of Unobserved States), wherein the sequencing dataset will be normalized and the metagenomes will be predicted against the PICRUSt-formatted, characterized- protein, functional database of Kyoto Encylopedia of Genes and Genomes (KEGG).

(10) TECHNOLOGY ROADMAP (if applicable) (use the attached sheet)



(11) EXPECTED OUTPUTS (6Ps)

6Ps	Year 1	Year 2
Publication	None	At least 1 publication in ISI/Scopus Journal At least 1 paper presentation in conferences
Patent/ Intellectual Property	None	At least 1 IP protection for the SMS Aquafeed
Product/ Process		One (1) product (SMS aquafeed)
People Services	Mentored/ Trained at least 2 researchers	Mentored/ Trained at least 2 researchers, 2 undergraduate students
Places and Partnership	None	MOA with UPLB, DA RFO5 and Bicol Mushroom Farmers
Policy	None	At least one (1) policy recommendation for potential use of SMS aquafeed in aquaculture farming.

(12) POTENTIAL OUTCOMES

1. Optimum inclusion rate of SMS in tilapia feeds
2. Gut metagenome profile of tilapia as affected by dietary inclusion of SMS
3. Characterized SMS aquafeed product for tilapia aquaculture.
4. Mentored and/or trained researchers in formulating and developing SMS aquafeed.

(13) POTENTIAL IMPACTS (2Is)

Social Impact
This project will help promote awareness to farmers and various stakeholders in using alternative ingredients such as SMS in formulating aquafeeds for tilapia. The use of SMS as an alternative aquafeed ingredient will offer alternatives in achieving higher-quality aquaculture. This will also help reduced the tons of SMS produced in the mushroom industry by recycling its generated wastes.

Economic Impact
The use of SMS as an alternative ingredient for aquafeed formulation can help reduced the feed costs and improve profit margins. This also promotes sustainable practices by using waste products generated by the mushroom industry.

(14) TARGET BENEFICIARIES

1. Local farmers engaged in aquaculture farming
2. LGUs and cooperatives
3. Feed manufacturers and compounders
4. Academe

(15) SUSTAINABILITY PLAN (if applicable)

This project can generate income by producing SMS aquafeeds for local farmers engaged in the aquaculture farming. Proceeds from this service will be used to sustain future project activities.

(16) GENDER AND DEVELOPMENT (GAD) SCORE (refer to the attached GAD checklist)

(17) LIMITATIONS OF THE PROJECT

The project will only use the popularly used substrate in mushroom cultivation. Feeding trial testing will only be conducted in tilapia in an aquarium tank set-up. The data that will be generated from this project will be used as a baseline data for future large- scale production and farm trial testing.

(18) LIST OF RISKS AND ASSUMPTIONS RISK MANAGEMENT PLAN (List possible risks and assumptions in attaining target outputs or objectives.)

Key Risks	**Action Plan to Prevent/ Address Risk**
Delay in the procurement of goods needed by the project.	Plan ahead of time to mitigate the delays in the procurement process. PPMP will be prepared ahead of time and specifications for each item will be checked for completeness.
Delay in the delivery of supplies and equipment.	Proper coordination with the supplier and formulation of catch-up plans to compensate for the delays.
Insufficient results. Insignificant data.	Additional tests and confirmatory experiments will be done.
Unavailability of chemical reagents and laboratory supplies.	Substitute reagents will be used and modifications of methods will be done.
Power failure/ fluctuations	Utilization of generator as back-up power supply.
Unexpected results of experimentation compared to theoretical data/ expected output.	Use of alternate methods and conduct of additional tests and experiments for failure-analysis factors.

(19) LITERATURE CITED

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(20) PERSONNEL REQUIREMENT

Position	Percent Time Devoted to the Project	Responsibilities
Project Leader (1)	20	The project leader must clearly define the project's goals, objectives, and scope to ensure that everyone involved understands the project's purpose and deliverables. It is responsible for creating a detailed project plan that outlines the timeline, milestones, budget, and resources required to complete the project. The project leader assigns tasks and responsibilities to team members based on their skills and expertise. They also ensure that each team member understands their role and

		responsibilities within the project.
Project Staff (3)	20	Project staff are responsible for contributing to the successful completion of a project by fulfilling their assigned tasks and responsibilities. Ensures that the project objectives are attained at given period of time. Provides expertise in the assigned tasks and responsibilities Performs the assigned individual tasks in the project at a given period of time. Facilitates the data gathering, processing and interpretation of the data.
Science Research Assistant (1)	100	Carry out experiments and research according to protocols laid out by primary researchers. Collect and log experimental data. Conduct statistical analyses of data sets. Prepare graphs and spreadsheets to portray results.
Laboratory Aide (2)	100	Assist in conducting tests and cleaning lab equipment, preparing media and maintaining the cleanliness of the laboratory.

(21) BUDGET BY IMPLEMENTING AGENCY

IMPLEMENTING AGENCY	PS	MOOE	EO	Total
Year 1	**987,852.00**	**1,430,738.90**	**0.00**	**2,418,590.90**
Year 2	**1,081,452.00**	**1,491,508.90**	**0.00**	**2,572,960.90**
TOTAL	**2,069,304.00**	**2,922,247.80**	**0.00**	**4,991,551.80**

(22) OTHER ONGOING PROJECTS BEING HANDLED BY THE PROJECT LEADER: 2 (number)

Title of the Project	Funding Agency	Involvement in the Project
Establishment of ParSU-MTL (Microbiology Testing Laboratory)	DOST-PCIEERRD	Project Leader
Nutrient Solubilization screening of Adiangao Cave- Dwelling Microbial Isolates	Partido State University,	Project Leader
Identification of Antibiotic-resistant bacteria and gut microbiome communities associated in fresh catch fish in Presentacion Coastal Area	Partido State University,	Project Leader

(23) OTHER SUPPORTING DOCUMENTS (Please refer to page 2 for the additional necessary documents.)

I hereby certify the truth of the foregoing and have no pending financial and/or technical obligations from the DOST and its attached Agencies. I further certify that the programs/projects being handled is within the prescribed number as stipulated in the DOST-GIA Guidelines. Any willful omission/false statement shall be a basis of disapproval and cancellation of the project.

	SUBMITTED BY (Project Leader)	ENDORSED BY (Head of the Agency)
Signature		
Printed Name	**LIEZEL DC. ATOLE, MSc, RMicro**	**RAUL G. BRADECINA, PhD**
Designation/Title	**Instructor 1/ Program Director**	**PSU President III**
Date		

Note: See guidelines/definitions at the back.

DOST Form 2 (for Basic/Applied Research)
DETAILED R & D PROJECT PROPOSAL

I. General Instruction: Submit through the DOST Project Management Information System (DPMIS), http://dpmis.dost.gov.ph, the detailed R&D proposal for the component project together with the detailed proposal of the whole Program, project workplan, line-item budget (LIB), 1-page curriculum vitae of the Project Leader, and Certificate of Incorporation or DTI Registration (if applicable) and other applicable supporting documents required under item II.23 below. Also, submit four (4) copies of the proposal together with its supporting documents. Use Arial font, 11 font size.

II. Operational Definition of Terms:

1. Title- the identification of the Program and the component projects.

Project- refers to the basic unit in the investigation of specific S&T problem/s with predetermined objective/s to be accomplished within a specific time frame.

Project Leader- refers to a project's principal researcher/implementer.

Project Duration- refers to the grant period or timeframe that covers the approved start and completion dates of the project, and the number of months the project will be implemented.

Implementing Agency- the primary organization involved in the execution of a program/project which can be a public or private entity

2. Cooperating Agency/ies- refers to the agency/ies that support/s the project by participating in its implementation as collaborator, co-grantor, committed adopter of resulting technology, or potential investor in technology development or through other similar means.

3. Site/s of Implementation- location/s where the project will be conducted. Indicate the barangay, municipality, district, province, region, and country.

4. Type of Research- indicates whether the project is basic or applied.
Basic research- is an experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundations of phenomena and observable facts, without any particular or specific application or use in view.
Applied research- is an investigation undertaken in order to utilize data/information gathered from fundamental/basic researches or to acquire new knowledge directed primarily towards a specific practical aim or objective with direct benefit to society.

5. R&D Priority Area and Program- based on the Harmonized National R&D Agenda 2017-2022, indicates which R&D agenda the project can be categorized in: Agriculture, Aquaculture and Natural Resources; Health; Industry, Energy, and Emerging Technology; Disaster Risk Reduction and Climate Change Adaptation; and Basic Research. Indicate also the specific Commodity/Sector, whether crops, livestock, forestry, agricultural resources or socio-economics; fisheries or aquatic resources; biotechnical, pharmaceutical, or health services; biotechnology, information technology, material science, photonics or space technology; industry, energy, utilities or infrastructure.

Sustainable Development Goal (SDG) Addressed- indicates which among the 17 SDGs adopted by the United Nations Members States are addressed by the project

6. Executive Summary- briefly discusses what the whole proposal is about

7. Introduction- a formally written declaration of the project and its idea and context to explain the goals and objectives to be reached and other relevant information that explains the need for the project and aims to describe the amount of work planned for implementation; refers to a simple explanation or depiction of the project that can be used as communication material.

7.1. Rationale- brief analysis of the problems identified related to the project
Significance- refers to the alignment to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development.

7.2. Scientific Basis- other scientific findings, conclusions or assumptions used as justification for the research
Theoretical Framework- the structure that summarizes concepts and theories that serve as basis for the data analysis and interpretation of the research data.

7.3. Objectives- statements of the general and specific purposes to address the problem areas of the project.

8. Review of Literature- refers to the following: (a) related researches that have been conducted, state-of-the-art or current technologies from which the project will take off; (b) scientific/technical merit; (c) results of related research conducted by the same Project Leader, if any; (d) Prior Art Search, and; (e) other relevant materials.

9. Methodology- discusses the following: (a) variables or parameters to be measured and evaluated or analyzed; (b) treatments to be used and their layout; (c) experimental procedures and design; (d) statistical analysis; (e) evaluation method and observations to be made, strategies for implementation (Conceptual/Analytical framework).

10. Technology Roadmap (if applicable)**-** a visual document that communicates the plan for technology. It is a flexible planning technique to support strategic and long-range planning by matching short- and long-term goals to specific technology solutions.

11. Expected Outputs (6Ps)- deliverables of the project based on the 6Ps metrics (Publication, Patent/Intellectual Property, Product, People Service, Place and Partnership, and Policy).

Publication- published aspect of the research, or the whole of it, in a scientific journal or conference proceeding for peer review, or in a popular form.
Patent/Intellectual Property- proprietary invention or scientific process for potential future profit.
Product- invention with a potential for commercialization.
People Service- people or groups of people, who receive technical knowledge and training.
Place and Partnership- linkage forged because of the study.
Policy- science-based policy crafted and adopted by the government or academe as a result of the study.

12. Potential Outcomes- refer to the result that the proponent hopes to deliver three (3) years after the successful completion of the project.

13. Potential Impacts

Social Impact- refers to the effect or influence of the project to the reinforcement of social ties and building of local communities.
Economic Impact- refers to the effect or influence of the project to the commercialization of its products and services, improvement of the competitiveness of the private sector, and local, regional, and national economic development.

14. Target Beneficiaries- refers to groups/persons who will be positively affected by the conduct of the project.

15. Sustainability plan- refers to the continuity of the project or how it shall be operated amidst financial, social, and environmental risks.

16. Gender and Development (GAD) Score- refers to the result of accomplishing GAD checklists (for project monitoring and evaluation/project management and implementation) to highlight the contribution of the project in the achievement of the objectives of Republic Act 7192, “Women in Development and Nation Building Act,” interpreted as gender-responsive, gender-sensitive, has promising GAD concepts, or GAD is invisible.

17. Limitations of the Project- refer to restrictions or constraints in the conduct of the project.

18. Risk- refers to an uncertain event or condition that its occurrence has a negative effect on the project.
Assumption- refers to an event or circumstance that its occurrence will lead to the success of the project.

19. Literature Cited- an alphabetical list of reference materials (books, journals and others) reviewed. Use standard system for citation.

20. Personnel Requirement- details on the position of personnel to be involved in the project, percent time devoted to the project, and responsibilities.

21. Budget By Implementing Agency- personnel services (PS), maintenance and other operating expenses (MOOE), and equipment outlay (EO) requirement of the project by implementing agency for

Year 1 and for the whole duration of the project. Please refer to the DOST-GIA Guidelines for the details (Section IX.B of DOST Administrative Order (A.O.) 011, s. 2020).

a. **PS**- total requirement for wages, salaries, honoraria, additional hire and other personnel benefits.

b. **MOOE**- total requirement for supplies and materials, travel expenses, communication, and other services.

c. **EO**- total requirement for facilities and equipment needed by the Program.

22. Other Ongoing Projects Being Handled By the Project Leader- list of ongoing projects being handled by the Project Leader funded by the DOST-GIA Program and other sources, and the accompanying responsibilities relevant to the project.

23. Other supporting documents required- as stated in Section VII of DOST A.O. No. 011, Series of 2020 – Revised Guidelines for the Grants-in-Aid Program:

a. Detailed breakdown of the required fund assistance to indicate the counterpart of the proponent and other fund sources including letter/s of commitment from the implementing, collaborating and coordinating agency/entity/ies;[1]
b. A counterpart fund, in kind and/or in cash, shall be required from the implementing agency/entity as one of the application requirements. All projects must have a minimum of 15% counterpart contribution except for projects involving public good;[1]
c. Curriculum Vitae or Personal Data Sheet (PDS) of Project Leader and other co-researchers/implementers. The service record may be requested if needed;[1]
d. Clearance from the DOST or the Funding Agency (e.g., DOST Councils) on previously funded completed projects handled by the Project Leader;[1]
e. Approval from the institution's ethics review board for research involving human subjects or in the case of animal subjects, approval from the Bureau of Animal Industry (BAI) (for PCAARRD- and PCHRD-monitored projects);
f. Clearance from the DOST Biosafety Committee (DOST-BC) shall be required for research proposals involving the use of GMOs under contained use (i.e., experiments done in laboratories, screen house, green house). For projects other than contained use, they shall be referred to the appropriate agency. The DOST Sectoral Councils, after determination as to whether or not the proposal has biosafety implications, shall endorse the same to the DOST-BC in accordance with the prescribed format under Annex 3 of the Philippine Biosafety Guidelines for Contained Use of Genetically Modified Organisms (series of 2014) (if applicable); and
g. For the private non-profit/non-government/people's organizations and startups:
    i. Up-to-date Securities and Exchange Commission (SEC) registration, or Department of Trade and Industry (DTI) registration, or Cooperative Development Authority (CDA) registration certificate, or other authenticated copy of latest Articles of Cooperation and other related legal documents;
    ii. Co-signers Statement (if applicable);
    iii. Copy of latest Income Tax Return;
    iv. Mayor's permit where the business is located;
    v. Audited Financial Statements for the past three (3) years preceding the date of project implementation or in case of those with operation of less than 3 years, for the years in operation and proof of previous implementation of similar projects (or in the case of startups, at least for one (1) year);
    vi. Document showing that NGO/PO has equity to 20 percent of the total project cost, which shall be in the form of labor, land for the project site, facilities, equipment and the like, to be used in the project;
    vii. Disclosure of other related business, if any;
    viii. List and/or photographs of similar projects previously completed, if any, indicating the source of funds for implementation;
    ix. Sworn affidavit of secretary of the NGO/PO that none of its incorporators, organizers, directors or officers is an agent of or related by consanguinity or affinity up to the fourth civil degree to the official of the agency authorized to process and/or approved the proposed MOA, and release of funds;
h. For CSOs, compliance to regulations as required by the General Appropriations Act (GAA) pertaining to fund transfers to Civil Society Organizations (CSOs); and
i. For foundations, DOST certification as accredited by the Science and Technology Foundation Unit

[1] *required of all proposals*

III. Criteria for Evaluation:

A. Criteria for Evaluating Proposals

Criterion	Definition
Relevance or Significance	Aligned to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development
Technical / Scientific Merit	Sound scientific basis to generate new knowledge or apply existing knowledge in an innovative manner
Budget Appropriateness	The proposed budget is commensurate to the proposed work plan and deliverables.
Competence of Proponent	Proponent's expertise is relevant to the proposal and with proven competence to implement, manage and complete R&D programs/projects within the approved duration and budget.

B. Governing Council / Board and EXECOM's Evaluation Criteria

Criteria	Indicators	Raw Score
A. Soundness of Proposal (20%)	R&D addresses relevant sectoral need (applicable to pressing concern)	5
	Solution provided is most effective (compared to other proposed solutions)	5
	Proposed budget is reasonable (project is not expensive vis-a-vis output)	5
	Work plan is doable in a given timeframe	5
B. Suitability of Output (30%)	R&D output is cost-effective (cost is competitive in relation to new or existing products or process)	5
	Has identified partners to adopt the technology (with letter of support from the head of the company)	5
	Output can be commercialized (through an existing manufacturer, spin-off or start-up company)	5
	R&D utilization is timely (output should not be overtaken by other solutions)	5
C. Significance of Outcome (30%)	Economic: increase in productivity, increase in income, new jobs generated, high return of investment (ROI)	5
	Social: working partnerships established, training opportunities provided, policies adopted, increased access to basic services (i.e., food, health, education); political, cultural, gender sensitivity and inclusivity	5
	Environment: enhanced environmental health standards, no adverse effect to the environment	5
	Sustainability: sustainability mechanisms established in terms of institutional, financial and human resources capability (submission of a new proposal to sustain a completed or ongoing proposal does not constitute sustainability of the project)	5
D. Competence of Proponent (20%)	Proponent's expertise aligned with the proposal	5
	Collaboration with relevant agencies and/or industry partners	5
	Thorough understanding of the proposal's deliverables	5
	DOST has good experience with the proponent	5

C. Additional Criteria on Gender and Development (GAD)